

PRIMEFLOW CONSULTING SERVICES

Database Architecture & Replication Guide

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System Engine: SQLite 3 (WAL Mode Enabled)

SQL Configuration File: server/primeflow_database_setup.sql

1. Database Overview & File Paths

PrimeFlow uses an SQLite 3 relational database with Write-Ahead Logging (WAL) for high reliability, zero-configuration setup, and high concurrency.

File Locations Across Environments:

- Local Development: server/primeflow.db (plus primeflow.db-wal & primeflow.db-shm)
- Cloud / Render Server: /var/data/primeflow.db (Persistent Storage)
- Vercel / Serverless Environment: /tmp/primeflow.db

2. How to Access and Inspect the Database

Command Line (SQLite CLI):

```
$ sqlite3 server/primeflow.db
SQLite> .tables           -- List all database tables
SQLite> .schema users     -- Display table DDL schema
SQLite> SELECT id, name, email, role FROM users; -- Query registered users
```

Graphical User Interfaces (GUI Tools):

- DB Browser for SQLite (Free Open Source - sqlitebrowser.org)
- VS Code Extension: "SQLite Viewer" or "vscode-sqlite"
- DBeaver / Beekeeper Studio / TablePlus

3. Database Tables & Default Seed Accounts

Core Tables Summary:

1. users - User accounts, roles (client, admin, ops, compliance, supervisor)
2. applications - Client business filings (CAC Incorporation, BN, Compliance, etc.)
3. documents - Uploaded identity docs, CAC certificates, file approval status
4. messages - Consultation chat history between clients and staff officers
5. audit_logs - Security action logs, IP addresses, system audit trails
6. profiles - Company details, addresses, phone numbers, state/LGA
7. notifications - Real-time in-app alerts and notifications
8. verification_codes - OTP email registration codes
9. compliance_items - SCUML, Pencom, ITF, TCC, NSITF corporate compliance items

Default Seeded Accounts:

- System Admin: admin@primeflow.com (Password: admin123)
- Operations Officer: ops@primeflow.com (Password: ops123)
- Compliance Officer: compliance@primeflow.com (Password: compliance123)
- Corporate Client: client@primeflow.com (Password: client123)

DATABASE REPLICATION GUIDE

4. Step-by-Step Database Replication Instructions

Method A: Automatic Re-initialization & Seeding (Simplest)

1. Stop the backend Node server.
2. Delete primeflow.db, primeflow.db-wal, and primeflow.db-shm from the server/ directory:
`$ rm server/primeflow.db*`
3. Restart the server: `$ cd server && npm run dev`
4. The server automatically initializes all tables, indexes, and seeds test accounts.

Method B: Manual SQL Execution via .SQL File

1. Execute the standalone SQL file:
`$ sqlite3 server/primeflow.db < server/primeflow_database_setup.sql`
2. This creates all 9 tables, 10 performance indexes, and seeds initial system users.

Method C: Data Backup & Clone via SQL Dump

To create an exact duplicate of an active database including all client records:

1. Export dump: `$ sqlite3 server/primeflow.db .dump > backup_dump.sql`
2. Import dump: `$ sqlite3 server/replicated_primeflow.db < backup_dump.sql`

Method D: Production PostgreSQL / MySQL Migration

1. Datatype Conversions:
 - INTEGER PRIMARY KEY AUTOINCREMENT -> SERIAL PRIMARY KEY
 - TEXT -> TEXT / VARCHAR
 - TIMESTAMP DEFAULT CURRENT_TIMESTAMP -> TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
2. Run DDL queries on target server (Supabase / AWS RDS / Neon Postgres).